Edge Computing Architectures: Paradigms, Patterns, and Strategic Deployment

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Abstract

Edge computing relocates compute and storage from centralised cloud data centres to resources near the data source, addressing the latency, bandwidth, privacy, and regulatory limits that hyperscale cloud alone cannot satisfy [1], [2]. This article synthesises the architectural foundations of edge computing, compares the three dominant paradigms — cloudlets, fog computing, and Multi-access Edge Computing (MEC) — and examines modern cloud-native platforms, with a focus on lightweight Kubernetes distributions and serverless at the edge. It treats edge AI, federated learning, security under zero-trust assumptions, and data sovereignty, and maps representative application domains (autonomous driving, industrial IoT, smart cities, healthcare, immersive media) to the architectural variants that best serve them. The principal finding, consistent across the 2019–2025 survey literature, is that cloudlets, fog, and MEC have converged into a single end–edge–cloud continuum, and that the research frontier has shifted from *defining* the edge to *operating* it at scale [2], [6], [10]. The article closes with eight architectural principles, a six-tier reference architecture, and an evaluation checklist for architects.

Index Terms — Edge computing, fog computing, Multi-access Edge Computing, cloudlet, Kubernetes, federated learning, zero trust, end–edge–cloud continuum.

1. Introduction

Machine-generated data is growing faster than the capacity of backbone networks to carry it. Gartner estimated that by 2025 roughly 75 % of enterprise-generated data would be created and processed outside traditional data centres or cloud regions, compared with about 10 % in 2018 [30]. This *data-gravity* problem — large datasets are expensive and slow to move — has driven a structural shift: rather than transporting every byte to a hyperscale cloud, organisations push compute toward the data [1], [2], [11].

Edge computing is the resulting distributed paradigm. Shi *et al.* defined it as "enabling technologies that allow computation to be performed at the edge of the network, on downstream data on behalf of cloud services and upstream data on behalf of IoT services" [1]. Satyanarayanan later enumerated four motivations — latency, bandwidth, privacy as an "exposure firewall", and survivability — that remain the canonical justification for edge deployment [2].

The field has nevertheless fragmented terminologically. *Cloudlet*, *fog*, *MEC*, *micro data centre*, *mist*, and *CDN edge* emerged from different research and industrial communities, and authors frequently use them inconsistently. Three guiding questions organise this article:

- **RQ1.** What architectural variants of edge computing are distinguishable, and how are they related?
- **RQ2.** What design challenges dominate edge architecture, and how are they addressed?
- **RQ3.** Which architectural variants are best matched to which application classes?

The remainder of the article is structured as follows. Section 2 traces the evolution from centralised cloud to edge. Section 3 fixes a vocabulary and introduces the end–edge–cloud continuum. Section 4 presents the three foundational paradigms. Section 5 enumerates common deployment patterns. Sections 6–8 cover modern platforms, edge AI and federated learning, and security. Section 9 catalogues recurring challenges; Section 10 distils architectural principles. Section 11 maps use cases to architectures, and Section 12 presents a reference architecture and an evaluation checklist. Section 13 discusses the research frontier, Section 14 acknowledges limitations, and Section 15 concludes.

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2. From Cloud to the Edge

Computing has oscillated between centralisation and distribution for six decades: mainframes (1960s–70s), personal computing (1980s), client–server and the Internet (1990s), and cloud re-centralisation (2006–2015). The cloud exploited economies of scale but imposed round-trip latencies of tens to hundreds of milliseconds — unacceptable for autonomous vehicles, remote robotic surgery, and immersive AR/VR [1], [2].

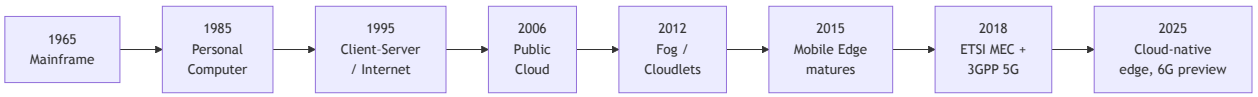

Figure 1. Evolution of dominant computing paradigms [1]–[5].

The 2012–2017 window is pivotal. Bonomi *et al.* coined *fog computing* as an IoT-oriented cloud-to-thing continuum [4]. Satyanarayanan *et al.* had already introduced *cloudlets* as trusted, resource-rich middle-tier nodes one wireless hop from the mobile device [3]. ETSI's Industry Specification Group standardised MEC as the telecommunications-native flavour of the edge, explicitly co-designed with 5G New Radio [7], [9]. The 2020s then reframed these three paradigms as *deployment flavours* of a shared end–edge–cloud continuum rather than competing alternatives [2], [10].

Edge architecture exists because some applications cannot tolerate a purely centralised control loop. Six drivers recur: **latency** for industrial control, robotics, and clinical workflows that need millisecond responses; **bandwidth** for video analytics and telemetry that exceed economical backhaul; **data locality and privacy** for healthcare, public sector, and manufacturing that must keep sensitive data within a site or jurisdiction; **resilience** for mines, ships, factories, and hospitals that cannot assume perfect WAN connectivity; **context** — decisions that need fresh local state (radio conditions, shop-floor state, traffic density); and **operational scale** — fleets of

thousands to millions of sites that demand automated deployment, identity, observability, and lifecycle management across heterogeneous hardware.

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3. Terminology and the End–Edge–Cloud Continuum

Architectural discussions often fail because teams use "edge" for different locations. The vocabulary below separates location, ownership, and function.

Term	Typical location	Architectural role
Device edge	Sensor, camera, PLC, vehicle ECU, robot, medical device	Capture data, run embedded inference, perform immediate safety or control functions
Gateway edge	Industrial PC, local gateway, router, appliance	Aggregate devices, normalise protocols, filter streams, enforce local policy
On-premises edge	Factory, hospital, store, branch, stadium, campus	Run site applications, local databases, video analytics, control-plane proxies
Access / network edge	5G RAN edge, central office, cable headend, ISP facility	Host latency-sensitive applications near users; integrate with network exposure and traffic steering
Metro / regional edge	Metro data centre or regional cloud zone	Place workloads close to a population or market while retaining richer cloud operations
Central cloud	Hyperscale or private cloud region	Global control, model training, fleet management, analytics, long-term storage
Fog computing	Distributed hierarchy between things and cloud	Broader continuum model emphasising compute, storage, control, and networking across multiple intermediate tiers
MEC	Multi-access Edge Computing host	ETSI-standardised edge environment for applications near access networks, not limited to mobile access

Table 1. A layered vocabulary for edge architecture, adapted from architectural practice.

Modern deployments treat these tiers as complementary layers of one architecture rather than competing alternatives. Ren *et al.*'s survey synthesises them into a four-tier end–edge–cloud architecture and is the highest-value single reference for the comparison [10]. A connected-vehicle system, for example, may run deterministic control in the vehicle, cooperative perception in MEC at the base station, multi-cell aggregation at a metro edge, and model training and fleet analytics in the central cloud — all governed from one global control plane.

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4. Architectural Paradigms: Cloudlet, Fog, and MEC

4.1 Cloudlets

A cloudlet is a small data centre or cluster "well-connected to the Internet and available for use by nearby mobile devices" [3]. Cloudlets trade hyperscale elasticity for proximity: a single wireless hop separates client from server, enabling crisp interactive response for mobile

augmented reality, wearables, and latency-sensitive offload. The cloudlet originated the VM-synthesis model, later generalised to container-based deployment.

4.2 Fog Computing

Fog computing, codified by the OpenFog Consortium, standardised as **IEEE 1934-2018**, and given a complementary conceptual model in NIST SP 500-325, is a hierarchical, multi-layer architecture distributing compute, storage, and applications across tiers between endpoint and cloud [5], [6]. Fog excels in large-scale industrial IoT where thousands of sensors must be managed across complex facilities with east–west communication between peer nodes [26].

4.3 Multi-access Edge Computing (MEC)

MEC, standardised by ETSI ISG MEC since 2014, places compute inside or adjacent to the Radio Access Network [7], [9]. *ETSI GS MEC 003 V4.1.1* (2025) defines the MEC host, platform, and orchestrator [7]. From 3GPP Release 17 onward, MEC is integrated into the 5G core, enabling session-level traffic steering to local edge applications [29]. MEC is the natural host for V2X, cooperative perception, and ultra-reliable low-latency communication (URLLC) workloads [8], [9], [22].

4.4 A Direct Comparison

Table 2. Comparison of the three dominant paradigms, synthesised from [6], [10].

Dimension	Cloudlet (CMU)	Fog (Cisco / OpenFog)	MEC (ETSI / 3GPP)
Deployment	Small DC, one wireless hop	Hierarchical device → gateway → regional	Inside RAN / base station / telco edge
Owner	Local business, campus	Enterprise / utility	Mobile network operator
Connectivity	Wi-Fi / WLAN	Any (WLAN, LTE, industrial bus)	Cellular (4G / 5G / 6G)
Latency target	10 – 50 ms	10 – 100 ms	**1 – 10 ms**
Virtualisation	VM synthesis (originally)	Containers, VMs	NFV / containers standardised by ETSI
Standardisation	Academic / OpenEdge	IEEE 1934 (OpenFog)	ETSI ISG MEC, 3GPP Rel-17/18
Mobility	Hand-off between cloudlets	Limited	Native (follow-me workloads, EAS re-sel.)
Best-fit use cases	AR / MAR, wearables, mobile offload	IIoT, smart grid, smart city	V2X / ADAS, immersive media, network slicing

The remaining distinctions are primarily sociological and contextual rather than architectural: a practitioner choosing between "deploying MEC" and "deploying a cloudlet" is in most cases choosing between standards ecosystems, not fundamentally different system designs [2], [11].

5. Common Deployment Patterns

Real deployments instantiate several of the components above simultaneously, and four patterns recur in practice.

1. **Local-first edge.** Operational application logic runs at the site; the cloud manages the fleet and aggregates data. Common in factories, hospitals, mines, ships, and retail. *Main challenge:* consistency — stale policies and delayed updates require explicit rules.
2. **Network edge / MEC.** Workloads sit in access-network or metro locations. Strongly relevant for telecom operators, 5G applications, V2X, cloud gaming, content delivery, and enterprise private networks [7], [29]. *Main challenge:* multi-party coordination across application owner, operator, cloud provider, enterprise, and device owner.
3. **Hierarchical edge.** Multiple tiers — device, gateway, site, metro, region, cloud — each with distinct function. Typical in smart cities, energy grids, transportation, and large-scale retail. *Main design issue:* placement of real-time control, local aggregation, privacy filtering, and historical learning.
4. **Edge–cloud continuum.** Compute, caching, and storage are dynamically allocated across tiers by time, location, user density, network condition, or energy cost. Suitable for connected-vehicle fleets, public-safety services, and distributed AI. *Main challenge:* dynamic placement must still satisfy latency, privacy, cost, and resilience constraints.

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6. Modern Edge Platforms

6.1 Kubernetes at the Edge

Kubernetes is the de facto orchestrator for cloud-native workloads, but upstream Kubernetes is too heavy for constrained edge nodes. This has driven a family of **lightweight Kubernetes distributions** [12], [14].

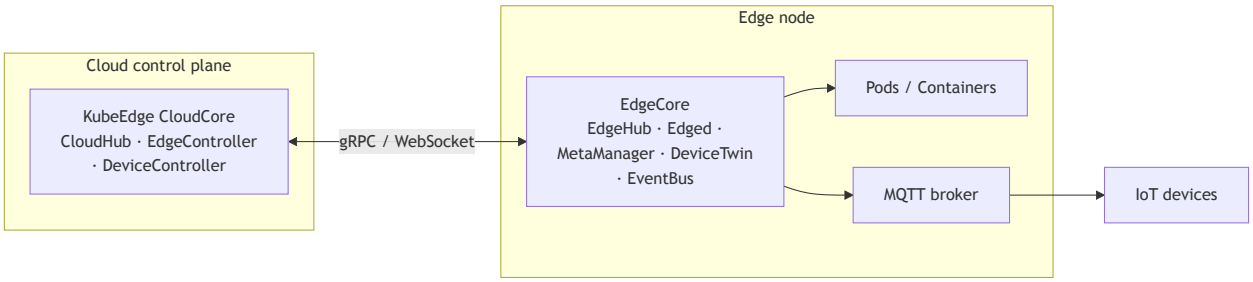

Figure 2. KubeEdge cloud–edge synchronisation architecture [13].

K3s, K0s, MicroK8s, KubeEdge, and OpenYurt each take a different path. Yakubov & Hästbacka's 2025 benchmark found K3s lowest on resources, K0s strongest on data-plane throughput, and KubeEdge richest in disconnected-operation features — at higher memory and disk-I/O cost [12]. Vaño *et al.*'s 2023 review reached compatible conclusions [14].

Table 3. Lightweight Kubernetes distributions for edge deployment (from [12], [14]).

Distribution	RAM footprint	Packaging	Disconnected-edge support	Notable trait
K3s	< 512 MB	Single binary	Limited	SQLite default DB; CNCF-certified
K0s	Minimal	Single binary	Limited	Best data-plane throughput
MicroK8s	~238 MB worker	Snap (Ubuntu)	Moderate	Rich addon system
KubeEdge	Higher (split)	CloudCore + EdgeCore	**Native**	Device twin, MQTT, offline pods
OpenYurt	Moderate	Non-intrusive to K8s	Good (YurtHub)	Zero delta to upstream kubelet

6.2 Serverless at the Edge

Function-as-a-Service (FaaS) extends the edge toward event-driven, ephemeral execution. Cloudflare Workers use V8 isolates; Fastly Compute uses WebAssembly on Wasmtime; AWS Lambda@Edge layers regional edge functions onto CloudFront [15]. Open-source stacks such as OpenFaaS, Knative, and Kubeless allow self-hosted FaaS on edge Kubernetes clusters. The principal operational concern remains **cold-start latency**, surveyed comprehensively by Golec *et al.* in 2024 with a taxonomy of mitigation techniques (pool-warming, snapshot resume, pre-initialised runtimes) [15].

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7. Edge AI and Federated Learning

7.1 Edge Inference and Hardware Acceleration

Zhou *et al.*'s canonical survey divides the field into "AI *for* the edge" (using AI to optimise orchestration) and "AI *on* the edge" (running inference on edge hardware) [17]. A mature edge platform integrates a heterogeneous silicon mix — CPU for orchestration, GPU or NPU for vision inference, FPGA for deterministic real-time processing — choosing per workload rather than per rack. Architecturally, edge AI demands model distribution and version control, hardware-aware placement, drift detection, rollback, and clear accountability for automated decisions.

7.2 Federated Learning

Federated learning (FL), introduced by McMahan *et al.* in 2017 as the FedAvg algorithm, trains global models without ever centralising raw data: nodes train locally and share only gradient updates with an aggregator [19]. Kairouz *et al.*'s 2021 agenda paper remains the canonical research reference [20]. Lim *et al.* bridged FL specifically to mobile edge networks [21].

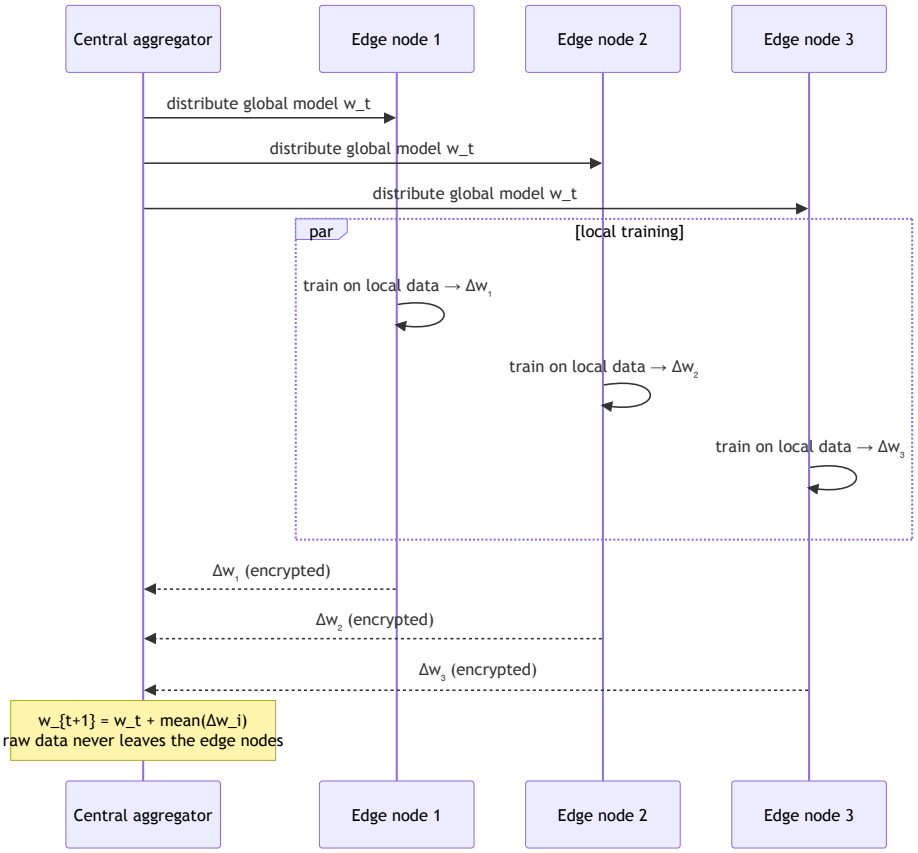

Figure 3. One round of federated averaging (FedAvg) [19].

With 6G targeting massive IoT, FL is becoming hierarchical: edge devices form clusters, aggregate locally at a base station or fog node, and only then contribute to a global update [16]. Letaief *et al.*'s 2022 vision paper is the canonical reference for 6G + edge AI convergence [16].

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8. Security, Governance, and Data Sovereignty

Edge deployments dramatically expand the attack surface: nodes are often physically accessible in uncontrolled environments. The **zero-trust (ZT) model** — "never trust, always verify" — is the foundational response [22]. ZT mandates continuous per-session verification (identity, device posture, location), microsegmentation, and distributed policy enforcement points that can isolate a compromised node in sub-second timescales without a central SOC. Sedjelmaci & Ansari (2024) add AI-augmented behavioural analysis tailored to 6G edge [23]. Confidential-computing primitives (Intel SGX/TDX, AMD SEV, ARM TrustZone/CCA, RISC-V Keystone) provide hardware-enforced Trusted Execution Environments with remote attestation, so a cloud or operator can verify an edge node's software state before provisioning secrets [24].

Table 4. Security and governance domains in edge architecture.

Domain	Architectural requirement	Example
Identity	Every device, node, workload, and operator has a verifiable identity	X.509 certificates for gateways, mutual TLS for service calls

Supply chain	Software and models are signed, traceable, and deployable by policy	Signed container images and signed ML-model artifacts
Physical risk	Nodes tolerate theft, tampering, and unplanned replacement	Encrypted disks, hardware-backed keys
Network exposure	Local services expose only required interfaces	Segmented OT network, controlled northbound data export
Data governance	Sensitive data is classified before movement	Medical images processed locally, metadata exported centrally
Observability	Security events survive intermittent links	Local buffering of audit logs with later forwarding to SIEM
Resilience	Sites degrade safely	Factory line continues with last approved model if WAN is down

Residency regulations (EU GDPR, HIPAA, DORA) require sensitive data to remain within specified jurisdictions. Edge architectures enable *partitioned multicloud*, *tiered hybrid*, and *air-gapped* sovereign patterns — crown-jewel data processed locally, elastic front-ends in public cloud — with hyperscalers offering dedicated sovereign zones (Microsoft Cloud for Sovereignty, AWS European Sovereign Cloud, Google Distributed Cloud air-gapped mode).

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9. Design Challenges

Three challenge clusters recur across every major survey.

- **Latency and bandwidth** justify edge placement. The Tactile Internet targets ~1 ms round-trip [34]; AR/VR motion-to-photon sits at ~20 ms; automotive V2X targets 3–10 ms per 3GPP specifications [29]. On the bandwidth side, HD video at 5 Mbps and 4K at 25 Mbps — multiplied across dense surveillance deployments — makes cloud-only analytics economically infeasible [32]. These numbers originate in standards and industry-requirements work rather than controlled experiments, and function as engineering targets rather than perceptual limits.
- **Orchestration, heterogeneity, and mobility** are the challenges *created* by edge computing. Uniform service deployment across x86, ARM, GPUs, FPGAs, and accelerators — plus session migration as users move — remains unresolved. ETSI MEC, Kubernetes-based frameworks (KubeEdge, OpenYurt), and cloud-provider edge extensions each solve a slice; no accepted standard yet spans multiple administrative domains [9], [35].
- **Security and privacy** change qualitatively at the edge: the attack surface is geographically distributed and often physically accessible. Roman, Lopez, and Mambo catalogued cross-paradigm threats — rogue edge nodes, man-in-the-middle, virtualisation escape, privacy leakage — and argued for defences portable across paradigms [33].

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10. Architectural Principles

Eight principles distil accumulated practice:

1. **Place workloads by constraint, not fashion.** Justify edge placement by latency, autonomy, bandwidth, privacy, process coupling, cost, or regulatory locality. If no constraint requires local placement, cloud placement is simpler and more reliable.
2. **Keep control loops local when safety or time requires it.** Closed-loop control should not depend on distant cloud round trips; cloud systems handle optimisation, learning, and coordination.
3. **Design for intermittent connectivity.** Define offline behaviour explicitly: what continues, what stops, what queues, what expires, what requires human approval.
4. **Minimise raw data movement.** Move events, aggregates, and exceptions; move raw data only for audit, training, or forensics.
5. **Treat edge sites as hostile or semi-trusted.** Secure boot, hardware roots of trust, disk encryption, measured updates, least privilege, certificate-based identity, short-lived credentials.
6. **Make operations fleet-oriented.** Desired-state management, staged rollouts, automated rollback, policy templates — every edge node replaceable without artisanal reconstruction.
7. **Separate data, control, and management planes.** Confusing them creates fragile systems and unclear security boundaries.
8. **Prefer open interfaces where ecosystems matter.** Standards reduce lock-in and integration risk in telecom MEC, industrial automation, smart cities, and connected-vehicle systems.

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11. Use Cases Mapped to Architectures

Edge computing's justification ultimately rests on specific workloads whose latency, bandwidth, privacy, or survivability requirements the public cloud cannot meet. Table 5 consolidates the mapping from application class to architectural variant.

Table 5. Use-case to architecture mapping, synthesised from [17], [18], [22], [25]–[29], [32].

Application class	Dominant constraint	Best-matched architecture	Representative references
Autonomous driving / V2X	3–10 ms latency	Hybrid: on-vehicle mist + RSU fog + MEC at base station	[25], [29]
AR / VR and cognitive assistance	<20 ms motion-to-photon	Cloudlet / MEC	[2], [28]
Industrial IoT / Industry 4.0	1–10 ms servo loops; determinism (TSN, OPC UA)	Hierarchical fog on-premises	[26]
Smart-city video analytics	100–500 ms soft real-time; bandwidth	Hierarchical fog + cloud training	[27], [32]
Smart healthcare	10 ms – 1 s; privacy	On-premises edge + federated learning	[20], [21]

Smart grid	Seconds (control); ms (protection)	Hierarchical fog	[4], [26]
Retail, hospitality, branch	Low-touch ops, WAN survivability	Local-first edge + cloud fleet mgmt	[31]
CDN-style content delivery	10–100 ms	Flat CDN edge + programmable FaaS	[15]
Immersive media / metaverse	Motion-to-photon; bandwidth	MEC + metro edge	[28], [29]

The most latency-constrained workloads (automotive, AR/VR, industrial control) require *hybrid* architectures combining extreme-edge and near-edge tiers — no single tier suffices. Soft-real-time workloads are adequately served by hierarchical fog or MEC alone. Bandwidth-dominant workloads such as video analytics should co-locate compute with the source regardless of latency profile.

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12. Reference Architecture and Evaluation Checklist

A general-purpose edge architecture is organised as: **(1) Device tier** — sensors, cameras, vehicles, medical devices; **(2) Gateway tier** — protocol normalisation, device identity, local buffering; **(3) Site edge tier** — local applications, stream processing, inference, site policy; **(4) Network / metro edge tier** — latency-sensitive shared services, MEC workloads, mobility support; **(5) Cloud tier** — global management, analytics, model training, governance; and **(6) Operations plane** (cross-cutting) — inventory, deployment, monitoring, security policy, and audit across all tiers. The reference is adapted by constraint: a retail store may need only gateway, site, and cloud; a connected-vehicle system needs device, roadside, network edge, regional, and cloud; a hospital emphasises on-premises autonomy over network-edge federation.

Before approving an edge architecture, an architect should be able to answer:

1. Which workload functions require local placement, and why?
2. What is the maximum acceptable latency for each control loop?
3. Which data must remain local, and for how long?
4. What continues during WAN outage, and what stops?
5. How are software and model updates signed, staged, verified, and rolled back?
6. How is observability preserved when links are intermittent?
7. How are edge nodes replaced without manual reconstruction?

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13. Discussion and Future Outlook

Three observations summarise the 2022–2025 research frontier.

First, paradigm convergence. Cloudlets, fog, and MEC are no longer competing definitions. Satyanarayanan [2] and Ren *et al.* [10] treat them as deployment flavours of one continuum, and

every major 2022–2025 survey — Kong 2023, Cruz 2022 **Second, operational scale has replaced definitional debate.** Dominant research themes are now Kubernetes-based orchestration for the edge [12], [14], serverless cold-start mitigation [15], 6G-native edge AI [16], and zero-trust / confidential computing [23], [24]. The open question is no longer "*what is the edge?*" but "*how do we operate 10 000 edge sites reliably and securely?*"

Third, implementation maturity diverges by application domain. Autonomous vehicles and immersive media map cleanly onto the standardised MEC paradigm with ETSI/3GPP-compliant prototypes [25], [28], [29], while industrial IoT remains firmly anchored in the fog tradition with TSN and OPC UA [26]. Healthcare and smart cities draw from both [27].

Cutting across all three, ML-inference workloads are reshaping edge design away from the VM-based cloudlet era toward container- and accelerator-centric platforms [17], [18] — a shift that will dominate the next five years. The strongest framing for a modern survey is therefore that **edge computing has entered a "cloud-native edge" era**, defined by portable Kubernetes workloads, ETSI-standardised APIs, pervasive AI, and zero-trust-by-design.

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14. Limitations

This article covers architectural concepts and influential surveys; it does not benchmark specific vendor implementations (AWS Wavelength, Azure Private MEC, Google Distributed Cloud Edge) or reproduce primary measurements. The corpus is skewed toward English-language venues indexed by IEEE, ACM, and Elsevier; work from Chinese CCF-A venues, industrial conferences, and operator research divisions is underrepresented. Standards cited here are living documents — ETSI GS MEC 003 reached V4.1.1 in 2025 and continues to evolve; readers should verify current versions. The specific latency thresholds (20 ms AR/VR, 10 ms V2X, 1 ms Tactile Internet) are engineering targets from standards work, not established perceptual or physical limits. Despite these caveats, the principal findings — paradigm convergence, orchestration as the remaining hard problem, and the use-case mapping — are consistent with independent conclusions of multiple high-impact surveys.

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15. Conclusion

Edge computing is not a replacement for the cloud but a structural complement that addresses the physical and economic limits of hyperscale centralisation. The three classical paradigms — cloudlets, fog, and MEC — have converged into a unified end–edge–cloud continuum. Modern deployments are cloud-native, built on lightweight Kubernetes distributions and increasingly augmented by serverless runtimes. Edge AI and federated learning are moving intelligence *and* model training to the periphery, while zero-trust and confidential-computing primitives respond to the expanded attack surface. From autonomous driving to smart manufacturing to healthcare, the pattern is consistent: workloads whose latency, bandwidth, or privacy constraints the cloud cannot satisfy migrate to the edge, and the remaining workloads stay where elasticity is cheapest. Successful architectures therefore do not choose between cloud

and edge — they place each workload on the right tier of the continuum: local enough to act, connected enough to learn, standardised enough to integrate, and secure enough to trust.

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